

Plant Disease Detection using MobileNetV2

Abstract

This project implements a deep learning-based system for automated plant leaf disease detection. Leveraging MobileNetV2, a lightweight convolutional neural network, the model classifies leaf images into different disease categories or identifies them as healthy. The goal is to provide farmers and researchers with an accessible and efficient tool for early disease detection.

Introduction

Plant diseases significantly impact agricultural productivity. Manual inspection is time-consuming, subjective, and requires expert knowledge. This project aims to address these issues using computer vision and deep learning. A custom dataset of leaf images was used to train and evaluate the model, with deployment in a user-friendly Streamlit web app.

Tools Used

- Python
- TensorFlow / Keras - MobileNetV2 (Transfer Learning)
- Streamlit for deployment
- Pyenv and virtual environments for dependency management

Steps Involved in Building the Project

1. Data Collection & Preprocessing: Leaf images were organised into train, validation, and test sets. Images were resized to 224x224 and normalised.
2. Model Building: MobileNetV2 (pre-trained on ImageNet) was used as a feature extractor. A custom classifier head was added.
3. Training: The classifier head was trained first, followed by fine-tuning the deeper layers with lower learning rates.
4. Evaluation: The model was tested on unseen data, achieving over 90% accuracy.
5. Deployment: A Streamlit app was developed, allowing users to upload or capture leaf images and receive predictions in real-time

Conclusion

The project demonstrates the effectiveness of transfer learning in plant disease detection. With high accuracy and lightweight architecture, the model is deployable on web and edge devices. Future improvements may include expanding the dataset to more plant species and diseases, integrating explainability features (heatmaps), and optimising the model for mobile deployment.